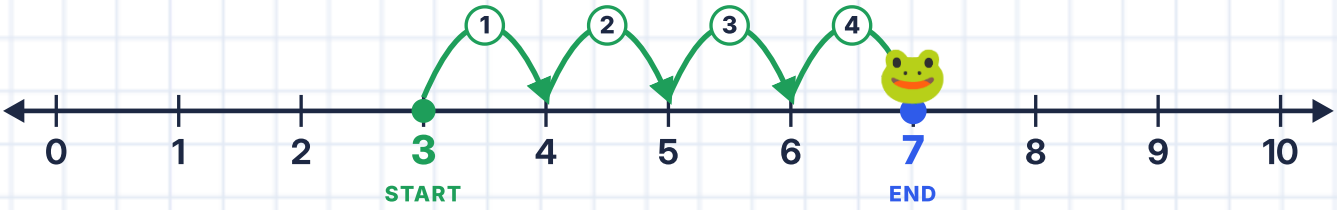
START HERE

Number Line Jumps

Meet **Hoppy the frog** 🐸. Hoppy will help you add and subtract by jumping on a number line.



IN THIS BOOK YOU WILL LEARN

1. What a number line is
2. Forward and backward
3. Adding: jump forward
4. Do not count the start!
5. Subtracting: jump backward
6. The sign is the boss
7. Start, jumps, end
8. Story clue words
9. Golden rules
10. Practice + answers

My name: _____ **Class:** _____

What is a number line?

A number line is like a **road of numbers**. The numbers stand in order.



LOOK CAREFULLY

← **Left side:** the **small** numbers (0, 1, 2 ...)

Right side →: the **big** numbers (... 8, 9, 10)

REMEMBER

Going RIGHT → numbers get BIGGER

Going LEFT ← numbers get SMALLER

YOUR TURN

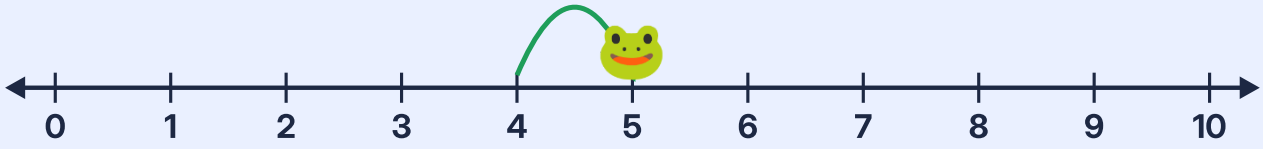
1. Which number is just after 6? _____ 2. Which number is just before 6? _____

3. Fill the missing numbers: 3, 4, _____, 6, _____, 8

Forward and backward

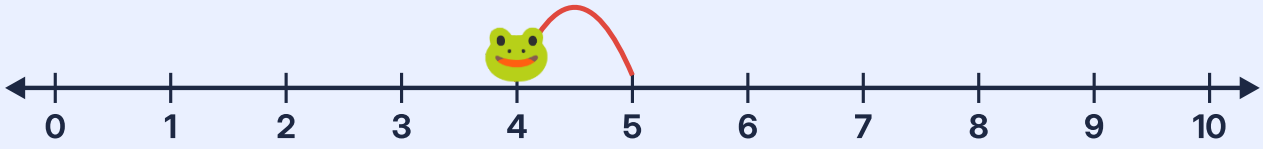
JUMPING FORWARD

Hoppy jumps to the **right →**. The number gets **bigger**. From 4 to 5.



JUMPING BACKWARD

Hoppy jumps to the **← left**. The number gets **smaller**. From 5 back to 4.



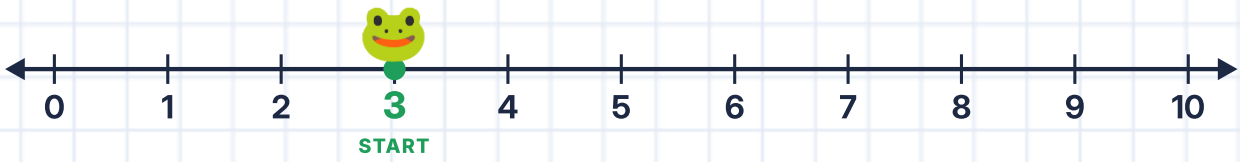
YOUR TURN

1. Hoppy is on 6. One jump forward. Hoppy lands on _____
2. Hoppy is on 6. One jump backward. Hoppy lands on _____
3. Hoppy is on 2. Two jumps forward. Hoppy lands on _____

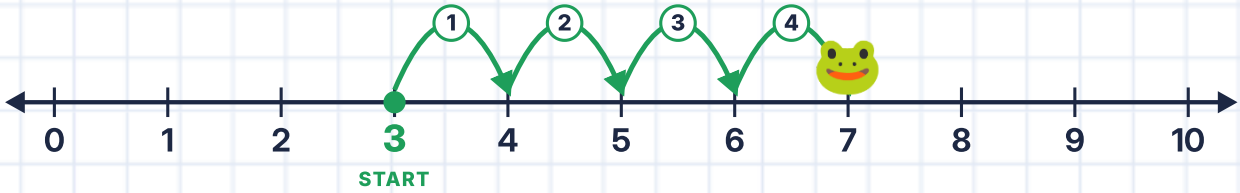
Adding: jump forward

Let us solve **3** + **4** = **?**. Plus means **more**. More means jump **forward →**.

1 **START on the FIRST number: 3. Hoppy just sits on 3. No jumping yet!**



2 **The SECOND number, 4, tells how many jumps. Jump 4 times forward. Count each jump when Hoppy lands: 1, 2, 3, 4.**



3 **Where did Hoppy land? On 7. That is the END and the answer.**

3 + 4 = 7

REMEMBER

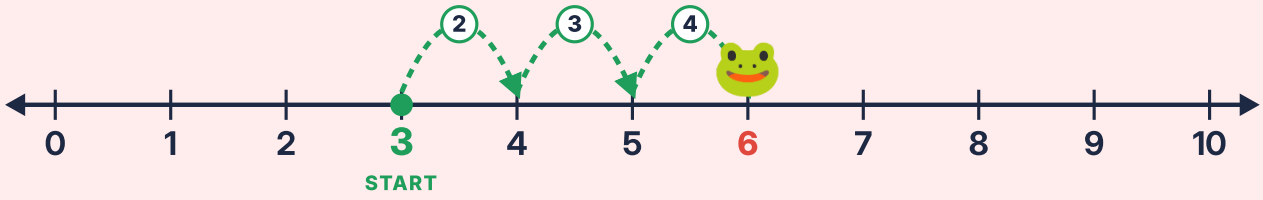
Start **3** · Jumps **4** · Way **forward →** · End **7**

Do not count the start!

This is the mistake most children make. Be a detective! 🔍

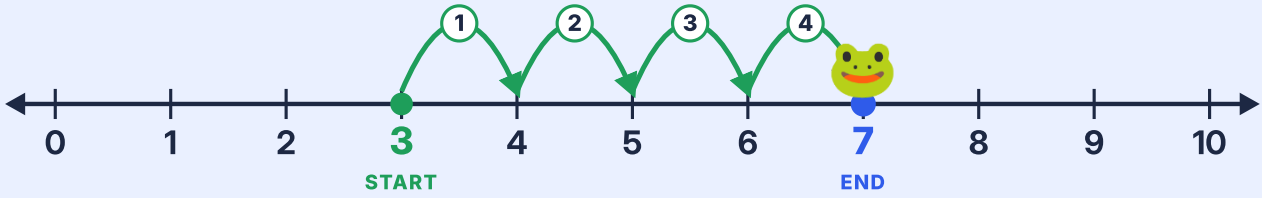
× WRONG WAY

A friend solved 3 + 4. They said "1" while **sitting** on 3. Then "2, 3, 4". They landed on **6**. Wrong!



✓ RIGHT WAY

Sitting is **not** jumping. The first jump goes **from** 3 and lands **on** 4. That is jump number 1.



FINGER TRICK

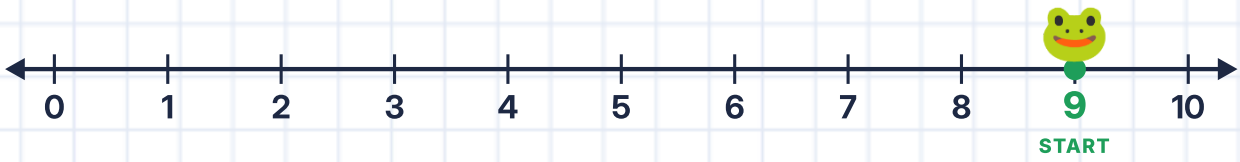
Put your finger on the start. Say **nothing**. Hop your finger and say "one" only when it **lands**.

Count the jumps, not the dots!

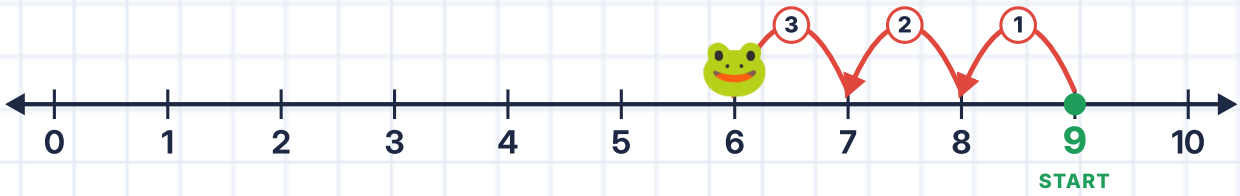
Subtracting: jump backward

Let us solve **9** **–** **3** **=** **?**. Minus means **take away**. Take away means jump **← backward**.

1 **START on the FIRST number: 9. (Not on 3!)**



2 **The second number is 3. Jump 3 times backward: 1, 2, 3.**



3 **Hoppy landed on 6. That is the answer.**

9 – 3 = 6

REMEMBER

For minus too, we start on the **first** number. The answer gets **smaller** because we went backward.

The sign is the boss

The sign in the middle tells Hoppy which way to go.

PLUS +



Jump **FORWARD**, to the right. The answer gets **bigger**.

MINUS −



Jump **BACKWARD**, to the left. The answer gets **smaller**.

YOUR TURN: WRITE F (FORWARD) OR B (BACKWARD)

5 + 2	_____	8 − 3	_____	6 + 4	_____
10 − 7	_____	2 + 9	_____	7 − 1	_____

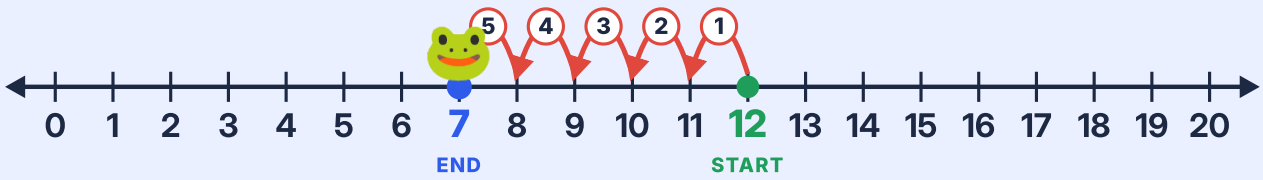
Start, jumps, end

Every number sentence gives each number a job.

6	+	3	=	9
START here	which WAY	how many JUMPS		END = answer

EXAMPLE: 12 − 5

Start on **12** · **backward** (minus) · **5** jumps · land on **7**.



12 − 5 = 7

YOUR TURN

15 − 4: Start _____ · Way _____ · Jumps _____ · End _____

8 + 6: Start _____ · Way _____ · Jumps _____ · End _____

Story clue words

Stories have no + or − sign. Look for **clue words**.

FORWARD → (ADD)

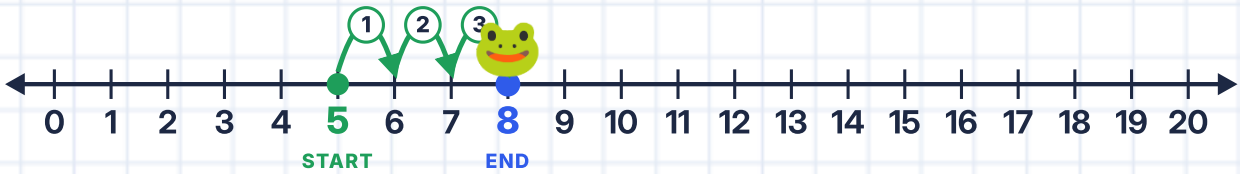
more · gets · gives her · joined · came · wins · in all · altogether

← BACKWARD (SUBTRACT)

gave away · ate · lost · flew away · left · fewer · spent · broke

1 **"Riya has 5 sweets. Mom gives her 3 more."**

"more" → **forward**. Start 5, 3 jumps → 8. **5 + 3 = 8**



2 **"There are 11 birds. 4 fly away. How many are left?"**

"fly away", "left" → **backward**. Start 11, 4 jumps → 7. **11 − 4 = 7**

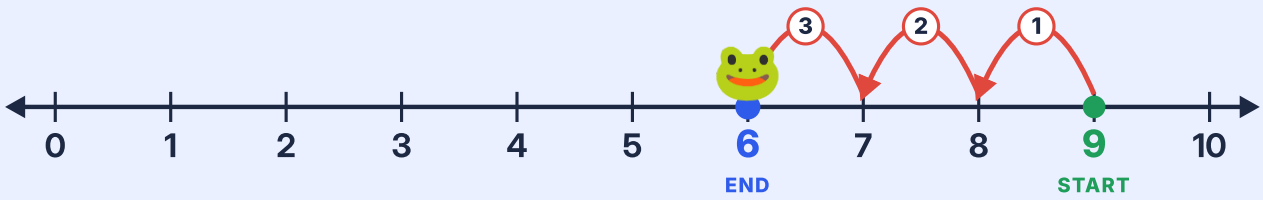


REMEMBER

Number line golden rules

1. **Start on the FIRST number.**
2. **Plus + → jump FORWARD (right).**
3. **Minus – → jump BACKWARD (left).**
4. **The SECOND number tells how many jumps.**
5. **Do not count the start. Count only the jumps.**
6. **Where Hoppy lands is the answer (END).**

EXAMPLE TO REMEMBER



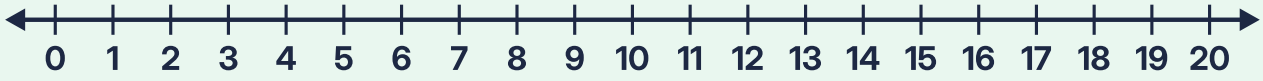
9 – 3: start 9 · backward · 3 jumps · end 6

PRACTICE

Your turn! (1)

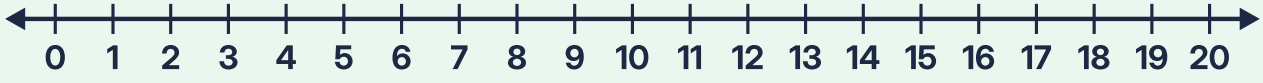
Draw the jumps on each number line. Mark the START and the END.

1. 5 + 3 = _____



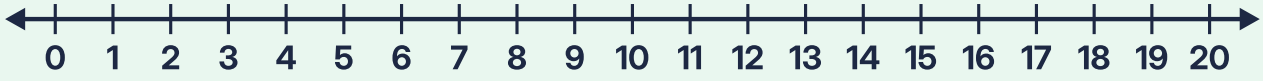
Start on _____ Way: forward / backward Jumps: _____ End: _____

2. 9 − 4 = _____



Start on _____ Way: forward / backward Jumps: _____ End: _____

3. 7 + 6 = _____

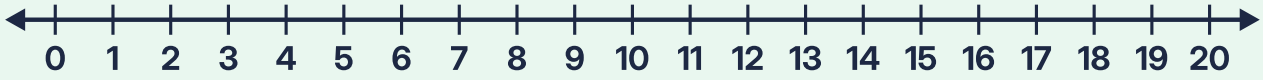


Start on _____ Way: forward / backward Jumps: _____ End: _____

PRACTICE

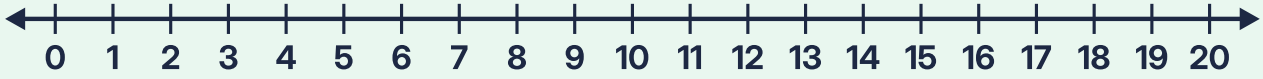
Your turn! (2)

4. 15 − 8 = _____



Start on _____ Way: forward / backward Jumps: _____ End: _____

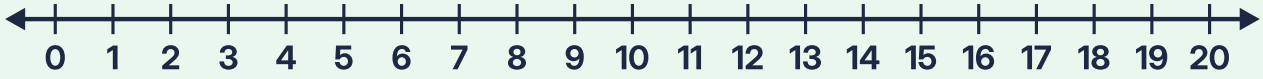
5. THERE ARE 12 APPLES. 5 ARE EATEN. HOW MANY ARE LEFT?



Start on _____ Way: forward / backward Jumps: _____ End: _____

Number sentence: _____

6. NEHA HAS 8 STICKERS. SHE GETS 9 MORE. HOW MANY NOW?



Start on _____ Way: forward / backward Jumps: _____ End: _____

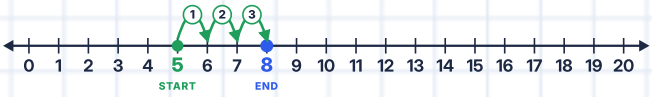
Number sentence: _____

Answers

Step 1: 7 · 5 · 5 and 7 **Step 2:** 7 · 5 · 4 **Step 6:** F, B, F, B, F, B

Step 7: 15 − 4: start 15, backward, 4 jumps, end 11 · 8 + 6: start 8, forward, 6 jumps, end 14

1. 5 + 3 = 8



2. 9 − 4 = 5



3. 7 + 6 = 13



4. 15 − 8 = 7



5. 12 − 5 = 7 APPLES LEFT



6. 8 + 9 = 17 STICKERS

